

APPENDIX - FOR ONLINE PUBLICATION

Appendix: Back to the 1980s or Not? The Drivers of Real and Inflation Risks in Treasury Bonds

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A Supply Side Microfoundations

A.1 Final good

A final consumption good is produced by a representative perfectly competitive firm from a continuum of differentiated goods $Y_{i,t}$:

$$Y_t = \left(Y_{i,t}^{\frac{\epsilon_p - 1}{\epsilon_p}} \right)^{\frac{\epsilon_p}{\epsilon_p - 1}}. \quad (\text{A1})$$

The constant $\epsilon_p > 1$ is the elasticity of substitution across intermediate goods. The resulting demand for the differentiated good i is downward-sloping in its product price $P_{i,t}$:

$$Y_{i,t} = Y_t \left(\frac{P_{i,t}}{P_t} \right)^{-\epsilon_p}. \quad (\text{A2})$$

The aggregate price level is given by

$$P_t = \left(\int_0^1 P_{i,t}^{-(\epsilon_p - 1)} di \right)^{-\frac{1}{\epsilon_p - 1}}. \quad (\text{A3})$$

A.2 Intermediate good producers

Intermediate goods firm i produces according to a Cobb-Douglas production function with constant returns to scale

$$Y_{i,t} = A_t N_{i,t}, \quad (\text{A4})$$

where productivity equals A_t and N_t is the supply of the aggregate labor index. Each firm takes the downward-sloping demand schedule as given (A3) and may therefore choose a different amount of the aggregate labor index. With the the final good equation (A1) aggregate output equals

$$Y_t = A_t N_t \quad (\text{A5})$$

where

$$N_t = \int_0^1 N_{i,t} di. \quad (\text{A6})$$

The aggregate resource constraint is simple because there is no real investment and consumption equals output:

$$C_t = Y_t. \quad (\text{A7})$$

Following [Lucas \(1988\)](#) we assume that productivity depends on past skills gained by all agents, and depends on past market labor, n_{t-1} :

$$a_t = \nu + a_{t-1} + (1 - \phi)n_{t-1}, \quad (\text{A8})$$

where $0 \leq \phi \leq 1$ and $\nu > 0$ are constants. The assumption [\(A8\)](#) ensures that potential output increases with past output. The process [\(A8\)](#) can equivalently be interpreted as a simple endogenous capital stock, similarly to [Woodford \(2003, Chapter 5\)](#), if a fixed proportion of market labor each period is used to produce investment goods with a constant-returns-to-scale technology, and the total amount of labor is scaled accordingly.

Intermediate firm i' real profit in period t equals

$$Pr_{i,t} = \frac{P_{i,t}}{P_t} Y_{i,t} - \frac{W_t}{P_t} N_{i,t}, \quad (\text{A9})$$

subject to the production function [\(A4\)](#), demand for differentiated goods [\(A2\)](#), and taking the wage W_t as given.

A.3 Employment agency

There is a continuum of monopolistically competitive households, each of which supplies a differentiated labor service, $L_{h,t}$, to the production sector. A representative employment agency aggregates households' labor hours according to a CES production technology with elasticity of substitution $\epsilon_w > 1$:

$$N_t = \left(\int_0^1 L_{h,t}^{\frac{\epsilon_w - 1}{\epsilon_w}} dh \right)^{\frac{\epsilon_w}{\epsilon_w - 1}} \quad (\text{A10})$$

The agency produces the aggregate labor index, N_t , taking each household's wage rate, $W_{h,t}$ as given, and then sells it to the production sector at the unit cost W_t . The profit maximization of the employment agency is:

$$\max_{L_{h,t}} W_t \left(\int_0^1 L_{h,t}^{\epsilon_w - 1} dh \right)^{\frac{1}{\epsilon_w}} - \int_0^1 W_{h,t} L_{h,t} dh, \quad (\text{A11})$$

which yields the following demand schedule for the labor hours of household h :

$$L_{h,t} = \left(\frac{W_{h,t}}{W_t} \right)^{-\epsilon_w} N_t. \quad (\text{A12})$$

The wage index faced by intermediary producers is then given by

$$W_t = \left(\int_0^1 W_{h,t}^{1-\epsilon_w} \right)^{\frac{1}{1-\epsilon_w}}. \quad (\text{A13})$$

A.4 Labor-leisure choice

Following the classic model of [Greenwood et al. \(1988\)](#), we assume that total consumption consists of a combination of market consumption and home production, given by:

$$C_{h,t}^{home} = A_t \left(1 - \frac{L_{h,t}^{1+\eta}}{1+\eta} \right) \quad (\text{A14})$$

Home production has decreasing returns to scale as in [Campbell and Ludvigson \(2001\)](#), and the parameter η determines the elasticity of market labor supply. Household h 's utility depends on market and home good consumption and the corresponding external habit levels H_t and H_t^{home} :

$$U_{h,t} = \frac{((C_{h,t} - H_t) + (C_{h,t}^{home} - H_t^{home}))^{1-\gamma} - 1}{1-\gamma} \quad (\text{A15})$$

We assume that home good habits are shaped by the aggregate consumption of home goods, so $H_t^{home} = C_t^{home}$ and in equilibrium home goods drop out of the utility function because all households end up choosing the same labor supply in equilibrium. Home production nonetheless matters for the wage-setting first-order condition, which depends on the marginal change in utility from choosing an off-equilibrium path labor supply. External market habit is described by the surplus consumption dynamics in the main paper.

A.5 Price- and wage-setting

We consider the simplified case with flexible product prices but sticky wages. Wage-setting frictions take the form of [Rotemberg \(1982\)](#). Specifically, we assume that wage-setters face a quadratic cost if they raise wages faster than past inflation. The indexing to past inflation is analogous to the indexing assumption in [Smets and Wouters \(2007\)](#) and [Christiano et al.](#)

(2005). The cost of re-setting wages for household h in terms of aggregate output equals

$$Cost^h = \frac{\gamma_w}{2} \left(\frac{W_{h,t}}{W_{h,t-1}} / \frac{W_{t-1}}{W_{t-2}} - 1 \right)^2 Y_t. \quad (\text{A16})$$

We assume that wage-setting costs get rebated to households lump-sum, i.e. aggregate consumption is unaffected.

A.6 Profit first-order condition

Because product prices are flexible, intermediate firm i 's profit becomes

$$Pr_{i,t} = Y_t \left(\left(\frac{P_{i,t}}{P_t} \right)^{-(\epsilon_p-1)} - \frac{W_t}{P_t A_t} \left(\frac{P_{i,t}}{P_t} \right)^{-\epsilon_p} \right). \quad (\text{A17})$$

Taking the first-order condition with respect to the relative price $\frac{P_{i,t}}{P_t}$ gives

$$\frac{P_{i,t}}{P_t} = \frac{\epsilon_p}{\epsilon_p - 1} \frac{W_t}{P_t A_t}. \quad (\text{A18})$$

Because in equilibrium all firms end up choosing the same price, we have that the real wage equals

$$\frac{W_t}{P_t} = \frac{\epsilon_p - 1}{\epsilon_p} A_t. \quad (\text{A19})$$

This means that due to partially monopolistic competition the real wage is compressed by a constant fraction relative to productivity and equilibrium profits of intermediary i are exactly proportional to aggregate output:

$$Pr_{i,t} = \frac{1}{\epsilon_p} Y_t. \quad (\text{A20})$$

This is good because a consumption claim is the same as a claim to firm profits.

A.7 Wage-setting first-order condition with flexible wages

To derive the wage-setting first-order condition, we first start by understanding what happens if wages are flexible. In this case, the first-order condition equals:

$$0 = \frac{d(C_{h,t} + C_{h,t}^{home})}{d(W_{h,t}/W_t)} \quad (\text{A21})$$

$$= \frac{d}{d(W_{h,t}/W_t)} \left[\frac{W_{h,t}}{P_t} \left(\frac{W_{h,t}}{W_t} \right)^{-\epsilon_w} N_t - \frac{A_t}{1+\eta} \left(\frac{W_{h,t}}{W_t} \right)^{-\epsilon_w(1+\eta)} N_t^{(1+\eta)} \right], \quad (\text{A22})$$

$$= \left[(-\epsilon_w + 1) \frac{W_t}{P_t} N_t \left(\frac{W_{h,t}}{W_t} \right)^{-\epsilon_w} + \epsilon_w A_t \left(\frac{W_{h,t}}{W_t} \right)^{-\epsilon_w(1+\eta)-1} N_t^{(1+\eta)} \right] \quad (\text{A23})$$

Because all wage-setters choose the same flexible-wage wage, we can set $W_{h,t} = W_t$. It then follows that the flexible-wage real wage increases proportionately with productivity and increases with the total amount of labor supplied

$$\frac{W_t^{flex}}{P_t} = \frac{\epsilon_w}{\epsilon_w - 1} A_t N_t^\eta, \quad (\text{A24})$$

$$= \frac{\epsilon_w}{\epsilon_w - 1} A_t^{1-\eta} Y_t^\eta \quad (\text{A25})$$

A.8 Sticky wage first-order condition

In the derivation of the wage Phillips curve we use the operator $\tilde{E}_t$ to denote the partially adaptive inflation expectations of wage-setters. With the quadratic wage-setting cost (A14) the first-order condition for wage-setting becomes

$$\begin{aligned} 0 = & \frac{d(C_{h,t} + C_{h,t}^{home})}{d\left(\frac{W_{h,t}}{W_t}\right)} - \gamma_w \left(\frac{W_{h,t}}{W_{h,t-1}} \frac{W_{t-2}}{W_{t-1}} - 1 \right) \frac{W_t}{W_{h,t-1}} \frac{W_{t-2}}{W_{t-1}} Y_t \\ & + \gamma_w \tilde{E}_t M_{h,t+1} \left(\frac{W_{h,t+1}}{W_{h,t}} \frac{W_{t-1}}{W_t} - 1 \right) \frac{W_{h,t+1}}{W_{h,t}} \frac{W_t}{W_{h,t}} \frac{W_{t-1}}{W_t} Y_{t+1}, \end{aligned} \quad (\text{A26})$$

Since there is symmetry (i.e. all households face the same problem), we can drop the h index when solving for the aggregate wage.

A.9 Log-linearizing the first-order wage-setting condition

Denoting the flexible wage steady-state output by $\bar{Y}_t$, we have that

$$\bar{Y}_t = A_t \bar{N}, \quad (\text{A27})$$

where the flexible-wage labor supply solves

$$\frac{\epsilon_p - 1}{\epsilon_p} = \frac{\epsilon_w}{\epsilon_w - 1} \bar{N}^\eta. \quad (\text{A28})$$

Using lower case for logs and hats to denote deviations from the flexible-wage equilibrium, the log output gap equals

$$x_t \equiv \hat{y}_t = n_t - \bar{n}, \quad (\text{A29})$$

$$= \hat{n}_t. \quad (\text{A30})$$

The steady-state stochastic discount factor equals

$$\bar{M}_{t,t+1} = \beta \exp(-\gamma g). \quad (\text{A31})$$

For convenience we define the constant

$$\beta_g \equiv \beta \exp(-(\gamma - 1)g). \quad (\text{A32})$$

Letting $\pi_t^w = \log \frac{W_t}{W_{t-1}}$ denote nominal log wage inflation and taking a first-order approximation around $\pi^w = 0$, expression (A26) simplifies to:

$$\begin{aligned} 0 = & (-\epsilon_w + 1) \frac{W_t}{P_t} N_t + \epsilon_w A_t N_t^{(1+\eta)} - \gamma_w (\pi_t^w - \pi_{t-1}^w) Y_t \\ & + \gamma_w Y_t \tilde{E}_t M_{t+1} (\pi_{t+1}^w - \pi_t^w) \frac{Y_{t+1}}{Y_t} \end{aligned} \quad (\text{A33})$$

Re-arranging:

$$\begin{aligned} \epsilon_w - 1 = & \epsilon_w A_t \frac{P_t}{W_t} N_t^\eta - \gamma_w (\pi_t^w - \pi_{t-1}^w) \frac{P_t}{W_t} \frac{Y_t}{N_t} \\ & + \beta \gamma_w \frac{P_t}{W_t} \frac{Y_t}{N_t} \tilde{E}_t M_{t+1} (\pi_{t+1}^w - \pi_t^w) \frac{Y_{t+1}}{Y_t}, \end{aligned} \quad (\text{A34})$$

$$\begin{aligned} = & \epsilon_w A_t \frac{P_t}{W_t} N_t^\eta - \gamma_w (\pi_t^w - \pi_{t-1}^w) \frac{P_t}{W_t} \frac{Y_t}{N_t} \\ & + \gamma_w \frac{P_t}{W_t} \frac{Y_t}{N_t} \beta_g \tilde{E}_t (\pi_{t+1}^w - \pi_t^w), \end{aligned} \quad (\text{A35})$$

where in the last step we dropped second-order terms in M_{t+1} and output growth interacted

with wage inflation. We next substitute the production function into (A35):

$$(\epsilon_w - 1) \frac{W_t}{A_t P_t} = \epsilon_w N_t^\eta - \gamma_w (\pi_t^w - \pi_{t-1}^w) + \beta^g \gamma_w \tilde{E}_t (\pi_{t+1}^w - \pi_t^w), \quad (\text{A36})$$

$$(\text{A37})$$

giving the wage Phillips curve

$$\pi_t^w = \frac{1}{1 + \beta_g} \pi_{t-1}^w + \frac{\beta^g}{1 + \beta_g} \tilde{E}_t \pi_{t+1}^w + \gamma_w^{-1} \left(\epsilon_w N_t^\eta - (\epsilon_w - 1) \frac{W_t}{A_t P_t} \right). \quad (\text{A38})$$

Note that the term in parentheses is the wedge between the real productivity-adjusted wage and workers' productivity-adjusted disutility of labor. Because we have flexible product prices the real productivity-adjusted wage is constant and we can substitute in from (A19):

$$\pi_t^w = \frac{1}{1 + \beta_g} \pi_{t-1}^w + \frac{\beta^g}{1 + \beta_g} \tilde{E}_t \pi_{t+1}^w + \gamma_w^{-1} \left(\epsilon_w N_t^\eta - (\epsilon_w - 1) \frac{\epsilon_p - 1}{\epsilon_p} \right) \quad (\text{A39})$$

In the flexible-wage equilibrium the term in parentheses is zero, giving the first-order log-linearization

$$\epsilon_w N_t^\eta - (\epsilon_w - 1) \frac{\epsilon_p - 1}{\epsilon_p} = \epsilon_w \bar{N}^\eta \exp(\eta \hat{n}_t) - (\epsilon_w - 1) \frac{\epsilon_p - 1}{\epsilon_p}, \quad (\text{A40})$$

$$\approx \epsilon_w \bar{N}^\eta \eta \hat{n}_t, \quad (\text{A41})$$

$$= \epsilon_w \bar{N}^\eta \eta \hat{y}_t \quad (\text{A42})$$

We therefore obtain the standard log-linearized wage Phillips curve

$$\pi_t^w = \frac{1}{1 + \beta_g} \pi_{t-1}^w + \frac{\beta^g}{1 + \beta_g} \tilde{E}_t \pi_{t+1}^w + \kappa \hat{y}_t, \quad (\text{A43})$$

where the constant κ equals

$$\kappa = \gamma_w^{-1} \epsilon_w \bar{N}^\eta \eta \quad (\text{A44})$$

Substituting in the adaptive inflation expectations assumption

$$\tilde{E}_t \pi_{t+1}^w = (1 - \zeta) E_t \pi_{t+1}^w + \zeta \pi_{t-1}^w, \quad (\text{A45})$$

gives the wage Phillips curve

$$\pi_t^w = \rho^\pi \pi_{t-1}^w + f^\pi E_t \pi_{t+1}^w + \kappa \hat{y}_t, \quad (\text{A46})$$

where

$$\rho^\pi = \frac{1}{1 + \beta_g} + \zeta - \frac{1}{1 + \beta_g} \zeta, \quad (\text{A47})$$

$$f^\pi = 1 - \rho^\pi. \quad (\text{A48})$$

Phillips curve shocks to (A46), $v_{\pi,t}$, arise from making the degree of monopolistic wage-setting frictions ϵ_w or the marginal cost of providing labor outside the home η time-varying.

A.10 Price inflation

Product prices equal

$$P_t = \frac{\epsilon_p}{\epsilon_p - 1} \frac{W_t}{A_t}, \quad (\text{A49})$$

so log price inflation equals (up to a constant)

$$\pi_t^p = \pi_t^w - \Delta a_t, \quad (\text{A50})$$

$$= \pi_t^w - (1 - \phi) \hat{y}_{t-1}, \quad (\text{A51})$$

where the log deviation of real GDP from potential is the output gap, i.e. $x_t = \hat{y}_t$.

B Solution

In the absence of demand shocks the lagged output gap does not enter as a separate state variable because x_{t-1} can be expressed as a linear combination of the time- t state vector Y_t . This is no longer possible in the presence demand shocks, thereby adding x_{t-1} as a new state variable for asset prices relative to [Campbell, Pflueger and Viceira \(2020\)](#).

B.1 Solving for macroeconomic dynamics

The full macroeconomic dynamics are determined by the Euler equation, the wage Phillips curve (A46) and the monetary policy rule, as well as the short-rate Fisher equation $r_t = i_t - E_t \pi_{t+1}^p$, the relationship between price and wage inflation (A51). The Euler equation is given by

$$x_t = f^x E_t x_{t+1} + \rho^x x_{t-1} - \psi (i_t - E_t \pi_{t+1}^p) + v_{x,t}, \quad (\text{A52})$$

where

$$\rho^x = \frac{\theta_2}{\phi - \theta_1}, \quad (\text{A53})$$

$$f^x = \frac{1}{\phi - \theta_1}, \quad (\text{A54})$$

$$\psi = \frac{1}{\gamma(\phi - \theta_1)}, \quad (\text{A55})$$

$$\theta_2 = \phi - 1 - \theta_1. \quad (\text{A56})$$

The wage Phillips curve is given by

$$\pi_t^w = \rho^\pi \pi_{t-1}^w + f^\pi E_t \pi_{t+1}^w + \kappa x_t + v_{\pi,t}, \quad (\text{A57})$$

The monetary policy rule is given by

$$i_t = \rho^i i_{t-1} + (1 - \rho^i) (\gamma^x x_t + \gamma^\pi \pi_t^p) + v_{i,t}, \quad (\text{A58})$$

where $v_{x,t} = \frac{1}{\gamma(\phi - \theta_1)} \xi_t$ denotes the demand shock; $v_{\pi,t}$ is the supply shock; and $v_{i,t}$ is the monetary policy shock.

We want to find a solution of the form

$$Y_t = BY_{t-1} + \Sigma v_t, \quad (\text{A59})$$

where the matrix B is $[3 \times 3]$, the matrix Σ is $[3 \times 3]$, and we work with the state vector

$$Y_t = [x_t, \pi_t^w, i_t]', \quad (\text{A60})$$

and the shock vector

$$v_t = [v_{x,t}, v_{\pi,t}, v_{i,t}]'. \quad (\text{A61})$$

Using the relationship (A51), we can write the macroeconomic dynamics in terms of the state vector Y_t :

$$Y_{1,t} = f^x E_t Y_{1,t+1} + \rho^x Y_{1,t-1} - \psi (Y_{3,t} - E_t Y_{2,t+1} + (1 - \phi) Y_{1,t}) + v_{x,t}, \quad (\text{A62})$$

$$Y_{2,t} = f^\pi E_t Y_{2,t+1} + \rho^\pi Y_{2,t-1} + \kappa Y_{1,t} + v_{\pi,t}, \quad (\text{A63})$$

$$Y_{3,t} = \rho^i Y_{3,t-1} + (1 - \rho^i) (\gamma^x Y_{1,t} + \gamma^\pi Y_{2,t} - \gamma^\pi (1 - \phi) Y_{1,t-1}) + v_{i,t}. \quad (\text{A64})$$

We can write this in matrix form:

$$0 = FE_t Y_{t+1} + GY_t + HY_{t-1} + Mv_t,$$

where the matrices F , G and H are given by

$$\begin{aligned} F &= \begin{bmatrix} \frac{f^x}{1+\psi(1-\phi)} & \frac{\psi}{1+\psi(1-\phi)} & 0 \\ 0 & f^\pi & 0 \\ 0 & 0 & 0 \end{bmatrix}, \\ G &= \begin{bmatrix} -1 & 0 & -\frac{\psi}{1+\psi(1-\phi)} \\ \kappa & -1 & 0 \\ (1-\rho^i)\gamma^x & (1-\rho^i)\gamma^\pi & -1 \end{bmatrix}, \\ H &= \begin{bmatrix} \frac{\rho^x}{1+\psi(1-\phi)} & 0 & 0 \\ 0 & \rho^\pi & 0 \\ -(1-\rho^i)(1-\phi)\gamma^\pi & 0 & \rho^i \end{bmatrix}. \end{aligned}$$

The matrix M is $[3 \times 3]$ and equals:

$$M = \begin{bmatrix} \frac{1}{1+\psi(1-\phi)} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{A65})$$

Following Uhlig (1999), we solve for the generalized eigenvectors and eigenvalues of the matrix Ξ with respect to the matrix Δ , where

$$\Xi = \begin{bmatrix} -G & -H \\ I_3 & 0_3 \end{bmatrix}, \quad (\text{A66})$$

$$\Delta = \begin{bmatrix} F & 0_3 \\ 0_3 & I_3 \end{bmatrix} \quad (\text{A67})$$

To obtain a solution, we then pick three generalized eigenvalues $\lambda_1, \lambda_2, \lambda_3$ with generalized eigenvectors $[\lambda z'_1, z'_1]'$, $[\lambda_2 z'_2, z'_2]'$, and $[\lambda_3 z'_3, z'_3]'$. We denote the diagonal matrix of these eigenvalues by $\Lambda = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, and the matrix of the lower $[3 \times 1]$ portion of the eigenvectors by $\Omega = [z_1, z_2, z_3]$. The corresponding solutions for B and Σ are then given by:

$$B = \Omega \Lambda \Omega^{-1}, \quad (\text{A68})$$

$$\Sigma = -[FB + G]^{-1} M. \quad (\text{A69})$$

For both our calibrations, there exist exactly three generalized eigenvalues with absolute value less than one, and we pick the non-explosive solution corresponding to these three eigenvalues.

B.2 Rotated state vector

Our state space for solving for asset prices is five-dimensional: It consists of $\tilde{Z}_t$, which a scaled version of Y_t , the surplus consumption ratio relative to steady-state $\hat{s}_t$, and the lagged output gap x_{t-1} .

We next describe the definition of $\tilde{Z}_t$. To simplify the numerical implementation of the asset pricing recursions, we require that shocks to the scaled state vector $\tilde{Z}_t$ are independent standard normal and that the first dimension of the scaled state vector is perfectly correlated with consumption innovations. This rotation facilitates the numerical analysis, because it is easier to integrate over independent random variables. Aligning the first dimension of the scaled state vector with output gap innovations (and hence surplus consumption innovations) helps, because it allows us to use a finer grid to integrate numerically over this crucial dimension over which asset prices are most non-linear.

If the scaled state vector equals $\tilde{Z}_t = AY_t$ for some invertible matrix A , the dynamics of $\tilde{Z}_t$ are given by:

$$\tilde{Z}_t = AY_t, \quad (\text{A70})$$

$$\tilde{Z}_{t+1} = \underbrace{ABA^{-1}}_{\tilde{B}} \tilde{Z}_t + \underbrace{A\Sigma v_{t+1}}_{\epsilon_{t+1}}. \quad (\text{A71})$$

We hence want a matrix, A , such that

$$\text{Var}(\epsilon_{t+1}) = A\Sigma\Sigma_v\Sigma' A', \quad (\text{A72})$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (\text{A73})$$

Finding such a matrix A should in general be possible, because the matrix M and therefore $\Sigma\Sigma_v\Sigma'$ generally have rank three. We require that the first dimension of ϵ_{t+1} is perfectly correlated with the consumption shock. We can therefore find the three rows of A using the following steps:

1. Set $A_1 = \frac{e_1}{\sqrt{e_1\Sigma\Sigma_v\Sigma'e_1'}}$.

2. We use the MATLAB function *null* to compute the null space of $A_1 \Sigma \Sigma_v \Sigma'$. Let n_2 denote the first vector in $\text{null}(A_1 \Sigma \Sigma_v \Sigma')$. We then define the second row of A as the normalized version of n_2 :

$$A_2 = \frac{n_2}{\sqrt{n_2 \Sigma \Sigma_v \Sigma' n_2}}. \quad (\text{A74})$$

3. Let n_3 denote the first vector in $\text{null}(A_1 \Sigma \Sigma_v \Sigma', A_2 \Sigma \Sigma_v \Sigma')$. We then define the third row of A as the normalized version of n_3 :

$$A_3 = \frac{n_3}{\sqrt{n_3 \Sigma \Sigma_v \Sigma' n_3}}. \quad (\text{A75})$$

It is then straightforward to verify that equation (A73) holds for

$$A = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix}. \quad (\text{A76})$$

B.3 Asset pricing recursions

Before deriving the recursions for the numerical asset pricing computations, we derive a convenient form for the dynamics of the log surplus consumption ratio. We use e_i to denote a row vector with 1 in position i and zeros elsewhere. The matrix

$$\Sigma_M = e_1 \Sigma \quad (\text{A77})$$

denotes the loading of consumption innovations onto the vector of shocks v_t , where e_1 is a basis vector with a one in the first position and zeros everywhere else. The volatility of consumption surprises equals:

$$\sigma_c^2 = \Sigma_M \Sigma_v \Sigma_M'. \quad (\text{A78})$$

To simplify notation, we define $\hat{s}_t$ as the log deviation of surplus consumption from its steady state. The dynamics of $\hat{s}_t$ are:

$$\hat{s}_t = s_t - \bar{s}, \quad (\text{A79})$$

$$\hat{s}_t = \theta_0 \hat{s}_{t-1} + \theta_1 x_{t-1} + \theta_2 x_{t-2} + \lambda(\hat{s}_{t-1}) \varepsilon_{c,t}, \quad (\text{A80})$$

where with an abuse of notation we write:

$$\lambda(\hat{s}_t) = \lambda_0 \sqrt{1 - 2\hat{s}_t} - 1, \hat{s}_t \leq s_{max} - \bar{s}, \quad (\text{A81})$$

$$\lambda(\hat{s}_t) = 0, \hat{s}_t \geq s_{max} - \bar{s}. \quad (\text{A82})$$

The steady-state surplus consumption sensitivity equals:

$$\lambda_0 = \frac{1}{\bar{S}}. \quad (\text{A83})$$

In our calculations of bond prices, we repeatedly substitute out expected log SDF growth, which equals:

$$E_t[m_{t+1}] = \log \beta - \gamma E_t \Delta \hat{s}_{t+1} - \gamma E_t \Delta c_{t+1}, \quad (\text{A84})$$

$$= -r_t + \xi_t - \frac{\gamma}{2}(1 - \theta_0)(1 - 2\hat{s}_t), \quad (\text{A85})$$

$$= -(e_3 - e_2 B + (1 - \phi)e_1)Y_t + \xi_t - \frac{\gamma}{2}(1 - \theta_0)(1 - 2\hat{s}_t) \quad (\text{A86})$$

We often combine this with $r_t = \bar{r} + (e_3 - e_2 B)Z_t$ and $\hat{r}_t = (e_3 - e_2 B)Z_t$.

Including the constant, consumption growth is given by:

$$\Delta c_{t+1} = g + x_{t+1} - \phi x_t. \quad (\text{A87})$$

The steady state real short-term interest rate at $x_t = 0$ and $s_t = \bar{s}$ is the same as in [Campbell and Cochrane \(1999\)](#):

$$\bar{r} = \gamma g - \frac{1}{2}\gamma^2 \sigma_c^2 / \bar{S}^2 - \log(\beta). \quad (\text{A88})$$

The updating rule for the log surplus consumption ratio can then be written in terms of the state variables as:

$$\hat{s}_{t+1} = \theta_0 \hat{s}_t + \theta_1 e_1 A^{-1} \tilde{Z}_t + \theta_2 x_{t-1} + \lambda(\hat{s}_t) \varepsilon_{c,t+1}. \quad (\text{A89})$$

B.3.1 Recursion for zero-coupon consumption claims

We now derive the recursion for zero-coupon consumption claims in terms of state variables $\tilde{Z}_t$, $\hat{s}_t$ and x_{t-1} . Let P_{nt}^c/C_t denote the price-dividend ratio of a zero-coupon claim on consumption at time $t + n$. The outline of our strategy here is that we first derive an analytic expression for the price-dividend ratio for P_{1t}^c/C_t . For $n \geq 1$ we guess and verify

recursively that there exists a function $F_n(\tilde{Z}_t, \hat{s}_t, x_{t-1})$, such that

$$\frac{P_{nt}^c}{C_t} = F_n(\tilde{Z}_t, \hat{s}_t, x_{t-1}). \quad (\text{A90})$$

The ex-dividend price-consumption ratio for a claim to all future consumption is then given by

$$\frac{P_t}{C_t} = F(\tilde{Z}_t, \hat{s}_t, x_{t-1}), \quad (\text{A91})$$

where we define

$$F(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = \sum_{n=1}^{\infty} F_n(\tilde{Z}_t, \hat{s}_t, x_{t-1}). \quad (\text{A92})$$

We now derive the recursion of zero-coupon consumption claims in terms of state variables $\tilde{Z}_t$ and $\hat{s}_t$. The one-period zero coupon price-consumption ratio solves:

$$\frac{P_{1,t}^c}{C_t} = E_t \left[\frac{M_{t+1}C_{t+1}}{C_t} \right] \quad (\text{A93})$$

We simplify

$$\begin{aligned} \frac{M_{t+1}C_{t+1}}{C_t} &= \beta \exp(-\gamma E_t \Delta \hat{s}_{t+1} - (\gamma - 1) E_t \Delta c_{t+1} \\ &\quad - \gamma(\hat{s}_{t+1} - E_t s_{t+1}) - (\gamma - 1)(c_{t+1} - E_t c_{t+1})). \end{aligned}$$

Using the notation $f_n = \log(F_n)$, this gives the log one-period price-consumption ratio as:

$$\begin{aligned} f_1(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \log \beta - \gamma[(\theta_0 - 1)\hat{s}_t + \theta_1 x_t + \theta_2 x_{t-1}] - (\gamma - 1)[g + E_t x_{t+1} - \phi x_t] \\ &\quad + \frac{1}{2}(\gamma \lambda(\hat{s}_t) + (\gamma - 1))^2 \sigma_c^2, \end{aligned} \quad (\text{A94})$$

$$\begin{aligned} &= \log \beta - (\gamma - 1)g - e_1[(\gamma \theta_1 - \gamma \phi + \phi)I + (\gamma - 1)B]A^{-1}\tilde{Z}_t \\ &\quad - \gamma(\theta_0 - 1)\hat{s}_t - \gamma \theta_2 x_{t-1} + \frac{1}{2}(\gamma \lambda(\hat{s}_t) + (\gamma - 1))^2 \sigma_c^2 \end{aligned} \quad (\text{A95})$$

Next, we solve for f_n , $n \geq 2$ iteratively. Note that:

$$\frac{P_{nt}^c}{C_t} = \mathbb{E}_t \left[\frac{M_{t+1}C_{t+1}}{C_t} \frac{P_{n-1,t+1}^c}{C_{t+1}} \right] = \mathbb{E}_t \left[\frac{M_{t+1}C_{t+1}}{C_t} F_{n-1}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right] \quad (\text{A96})$$

This gives the following expression for f_n :

$$f_n(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = \log \left[\mathbb{E}_t \left[\exp \left(\log \beta - (\gamma - 1)g - e_1[(\gamma\theta_1 - \gamma\phi + \phi)I + (\gamma - 1)B]A^{-1}\tilde{Z}_t - \gamma(\theta_0 - 1)\hat{s}_t - \gamma\theta_2 x_{t-1} - (\gamma(1 + \lambda(\hat{s}_t)) - 1)\sigma_c \epsilon_{1,t+1} + f_{n-1}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right) \right] \right]. \quad (\text{A97})$$

Here, $\epsilon_{1,t+1}$ denotes the first dimension of the shock ϵ_{t+1} .

B.3.2 Recursion for zero-coupon bond prices

We use $P_{n,t}^\$$ and $P_{n,t}$ to denote the prices of nominal and real n -period zero-coupon bonds. The strategy is to develop analytic expressions for one- and two-period bond prices. We then guess and verify recursively that the prices of real and nominal zero-coupon bonds with maturity $n \geq 2$ can be written in the following form:

$$P_{n,t} = B_n(\tilde{Z}_t, \hat{s}_t, x_{t-1}), \quad (\text{A98})$$

$$P_{n,t}^\$ = B_n^\$(\tilde{Z}_t, \hat{s}_t, x_{t-1}), \quad (\text{A99})$$

where $B_n(\tilde{Z}_t, \hat{s}_t, x_{t-1})$ and $B_n^\$(\tilde{Z}_t, \hat{s}_t, x_{t-1})$ are functions of the state variables. As discussed in the main paper, we assume that the short-term nominal interest rate contains no risk premium, so the one-period log nominal interest rate equals $i_t = r_t + E_t \pi_{t+1}$. Taking account of the constants, one-period bond prices equal:

$$P_{1,t}^\$ = \exp(-Y_{3,t} - \bar{r}), \quad (\text{A100})$$

$$P_{1,t} = \exp(-Y_{3,t} + \mathbb{E}_t Y_{2,t+1} - (1 - \phi)Y_{1,t} - \bar{r}). \quad (\text{A101})$$

We next solve for longer-term bond prices including risk premia. Substituting in (A100) into the bond-pricing recursion gives:

$$P_{2,t}^\$ = \exp(-\xi_t) \mathbb{E}_t [M_{t+1} P_{1,t+1}^\$ \exp(-Y_{2,t+1} + (1 - \phi)Y_{1,t})] \quad (\text{A102})$$

$$= \exp(-\xi_t) \mathbb{E}_t [M_{t+1} \exp(-Y_{3,t+1} - Y_{2,t+1} + (1 - \phi)Y_{1,t} - \bar{r})]. \quad (\text{A103})$$

We can now verify that the two-period nominal bond price takes the form (A99):

$$\begin{aligned}
B_2^{\$}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \exp(E_t(m_{t+1} - \xi_t - Y_{3,t+1} - Y_{2,t+1} + (1 - \phi)Y_{1,t}) - \bar{r}) \\
&\quad \times \mathbb{E}_t \left[\exp \left(\left(-\gamma(\lambda(\hat{s}_t) + 1) \Sigma_M - \underbrace{[(e_2 + e_3)\Sigma]}_{v_{\$}} \right) v_{t+1} \right) \right].
\end{aligned} \tag{A104}$$

Here, we define the vector $v_{\$}$ to simplify notation. Taking logs, substituting out for $E_t m_{t+1}$, and using the definition for the sensitivity function $\lambda(\hat{s}_t)$, we get:

$$\begin{aligned}
b_2^{\$} &= -e_3[I + B]A^{-1}\tilde{Z}_t + \frac{1}{2}v_{\$}\Sigma_v v_{\$}' \\
&\quad + \gamma(\lambda(\hat{s}_t) + 1)\Sigma_M \Sigma_v v_{\$}' - 2\bar{r}.
\end{aligned} \tag{A105}$$

The closed-form solution for the two-period real bond price becomes

$$\begin{aligned}
P_{2,t} &= \exp(E_t(m_{t+1} - \xi_t - Y_{3,t+1} - (1 - \phi)Y_{1,t+1} + Y_{2,t+2}) - \bar{r}) \\
&\quad \times \mathbb{E}_t \left[\exp \left(\left(-\gamma(\lambda(\hat{s}_t) + 1)\Sigma_M - \underbrace{(e_3 + (1 - \phi)e_1 - e_2 B)\Sigma}_{v_r} \right) v_{t+1} \right) \right]
\end{aligned} \tag{A106}$$

We define the vector v_r to simplify notation. Taking logs, substituting out for $E_t m_{t+1}$, and using the definition for $\lambda(\hat{s}_t)$ gives:

$$\begin{aligned}
b_2(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= -v_r[I + B]A^{-1}\tilde{Z}_t + \frac{1}{2}v_r \Sigma_v v_r' + \gamma(\lambda(\hat{s}_t) + 1)\Sigma_M \Sigma_v v_r' - 2\bar{r}.
\end{aligned} \tag{A107}$$

For $n \geq 3$, we repeatedly substitute out for $E_t m_{t+1}$ to obtain the following recursion for nominal and real bond prices, respectively:

$$\begin{aligned}
B_n^{\$}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \mathbb{E}_t \left[\exp \left(m_{t+1} - \xi_t - Y_{2,t+1} + (1 - \phi)Y_{1,t} + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, B^{\$}x_t) \right) \right] \\
&= \mathbb{E}_t \left[\exp \left(-\bar{r} - e_3 A^{-1} \tilde{Z}_t - \frac{\gamma}{2}(1 - \theta_0)(1 - 2\hat{s}_t) \right. \right. \\
&\quad \left. \left. - \gamma(1 + \lambda(\hat{s}_t))\sigma_c \epsilon_{1,t+1} - e_2 A^{-1} \epsilon_{t+1} + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right) \right].
\end{aligned} \tag{A108}$$

The value function iteration for real bond prices then becomes

$$\begin{aligned}
B_n(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \mathbb{E}_t \left[\exp \left(m_{t+1} - \xi_t + b_{n-1}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right) \right] \\
&= \mathbb{E}_t \left[\exp \left(-\bar{r} - (e_3 - e_2 B + (1 - \phi)e_1)A^{-1}\tilde{Z}_t - \frac{\gamma}{2}(1 - \theta_0)(1 - 2\hat{s}_t) \right. \right. \\
&\quad \left. \left. - \gamma(1 + \lambda(\hat{s}_t))\sigma_c \epsilon_{1,t+1} + b_{n-1}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right) \right]. \tag{A109}
\end{aligned}$$

B.3.3 Computing returns

The log return on the consumption claim equals:

$$r_{t+1}^c = \log \left(\frac{P_{t+1}^c + C_{t+1}}{P_t^c} \right), \tag{A110}$$

$$= \Delta c_{t+1} + \log \left(\frac{1 + \frac{P_{t+1}^c}{C_{t+1}}}{\frac{P_t^c}{C_t}} \right). \tag{A111}$$

Real and nominal log bond yields equal:

$$y_{n,t} = -\frac{1}{n}b_{n,t}, \tag{A112}$$

$$y_{n,t}^\$ = -\frac{1}{n}b_{n,t}^\$. \tag{A113}$$

Real log bond returns equal:

$$r_{n,t+1} = b_{n-1,t+1} - b_{n,t}. \tag{A114}$$

Nominal log bond returns equal:

$$r_{n,t+1}^\$ = b_{n-1,t+1}^\$ - b_{n,t}^\$. \tag{A115}$$

Real and nominal bond log excess returns then equal:

$$xr_{n,t+1} = r_{n,t+1} - r_t, \tag{A116}$$

$$xr_{n,t+1}^\$ = r_{n,t+1}^\$ - i_t. \tag{A117}$$

B.3.4 Levered stock prices and returns

We note that the price of the levered equity claim is δP_t^c , so the price-dividend ratio equals:

$$\frac{P_t^\delta}{D_t^\delta} = \delta \frac{C_t}{D_t^\delta} \frac{P_t^c}{C_t}. \quad (\text{A118})$$

Using the expression

$$D_{t+1}^\delta = P_{t+1}^c + C_{t+1} - (1 - \delta)P_t^c \exp(r_t) - \delta P_t^c, \quad (\text{A119})$$

and

$$P_t^\delta = \delta P_t^c \quad (\text{A120})$$

gives the gross return on levered stocks:

$$(1 + R_{t+1}^\delta) = \frac{D_{t+1}^\delta + P_{t+1}^\delta}{P_t^\delta}, \quad (\text{A121})$$

$$= \frac{1}{\delta} \frac{P_{t+1}^c + C_{t+1} - (1 - \delta)P_t^c \exp(r_t)}{P_t^c}, \quad (\text{A122})$$

$$= \frac{1}{\delta} (1 + R_{t+1}^c) - \frac{1 - \delta}{\delta} \exp(r_t). \quad (\text{A123})$$

Log stock excess returns then equal:

$$xr_{t+1}^\delta = r_{t+1}^\delta - r_t. \quad (\text{A124})$$

To mimic firms' dividend smoothing in the data, we report simulated moments for the price of equities dividend by dividends smoothed over the past 64 quarters:

$$P_t^\delta / \left(\frac{1}{64} (D_t^\delta + D_{t-1}^\delta + \dots + D_{t-63}^\delta) \right). \quad (\text{A125})$$

B.4 Risk-premium decomposition

We use the superscript rn for risk-neutral, superscript cf for cash flow, and rp for risk premium. Risk-neutral valuations are expected cash flows discounted with the risk-neutral discount factor, given by:

$$M_{t+1}^{rn} = \exp(-(r_t - \xi_t)). \quad (\text{A126})$$

Note that since we are not interested in risk-neutral bond and stock prices, but only a decomposition of returns, multiplying M_{t+1}^{rn} by a constant discount rate does not matter. For any zero-coupon claim it would shift risk-neutral returns merely by a constant and therefore leave our decomposition into risk-neutral and risk-premium components unaffected. For a claim to all future consumption or stock returns, a constant discount rate could theoretically shift the weights between nearer-term consumption claims and longer-term consumption claims, and therefore change risk-neutral returns. However, since consumption growth is stationary we have found that this makes very little difference to risk-neutral stock returns in any of our numerical applications.

B.4.1 Risk-neutral zero-coupon bond prices

We use analogous recursions to solve for risk-neutral bond prices. One-period risk-neutral bond prices are given exactly as before by equations (A100) and (A101). For $n > 1$, we guess and verify that the prices of real and nominal risk-neutral zero-coupon bonds with maturity n can be written in the following form

$$P_{n,t}^{rn} = B_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}), \quad (\text{A127})$$

$$P_{n,t}^{\$,rn} = B_n^{\$,rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}). \quad (\text{A128})$$

for some functions $B_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1})$ and $B_n^{\$,rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1})$.

We derive the two-period risk-neutral nominal bond price analytically:

$$P_{2,t}^{\$,rn} = \exp(-\xi_t) \mathbb{E}_t \left[M_{t+1}^{rn} P_{1,t+1}^{\$,rn} \exp(-Y_{2,t+1} + (1-\phi)Y_{1,t}) \right] \quad (\text{A129})$$

$$= \exp(-r_t) \mathbb{E}_t [\exp(-Y_{3,t+1} - Y_{2,t+1} + (1-\phi)Y_{1,t} - \bar{r})]. \quad (\text{A130})$$

We can hence verify that the two-period risk-neutral nominal bond price takes the form (A99)

$$b_2^{\$,rn} = -e_3 [I + B] A^{-1} \tilde{Z}_t + \frac{1}{2} v_{\$} \Sigma_v v_{\$}' - 2\bar{r} \quad (\text{A131})$$

Here, the vector $v_{\$}$ is identical to the case with risk aversion. Comparing expressions (A131) and (A105) shows that they agree when $\gamma = 0$. We similarly solve for 2-period real bond

prices in closed form:

$$P_{2,t}^{rn} = \exp(-Y_{3,t} + \mathbb{E}_t Y_{2,t+1} - (1-\phi)Y_{1,t} - \bar{r}) \times \exp(\mathbb{E}_t(-Y_{3,t+1} + \mathbb{E}_{t+1} Y_{2,t+2} + (1-\phi)Y_{1,t+1} - \bar{r})) \\ \times \mathbb{E}_t \left[\exp \left(- \underbrace{(e_3 + (1-\phi)e_1 - e_2 B) \Sigma v_{t+1}}_{v_r} \right) \right]. \quad (\text{A132})$$

The vector v_r is again identical to the case with risk aversion. Taking logs gives:

$$b_2^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = -(e_3 + (1-\phi)e_1 - e_2 B) [I + B] A^{-1} \tilde{Z}_t + \frac{1}{2} v_r \Sigma_v v_r' - 2\bar{r}. \quad (\text{A133})$$

We note that the risk-neutral bond prices (A133) and bond prices with risk aversion (A107) are identical when the utility curvature parameter γ equals zero.

For $n \geq 3$ the n -period risk neutral nominal and real bond prices satisfy the following recursions, respectively:

$$B_n^{\$,rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = \mathbb{E}_t \left[\exp \left(-\bar{r} - e_3 A^{-1} \tilde{Z}_t - e_2 A^{-1} \epsilon_{t+1} + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right) \right] \quad (\text{A134})$$

$$B_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = \mathbb{E}_t \left[\exp \left(-\bar{r} - (e_3 + (1-\phi)e_1 - e_2 B) A^{-1} \tilde{Z}_t + b_{n-1}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right) \right] \quad (\text{A135})$$

B.4.2 Risk-neutral zero-coupon consumption claims

Next, we derive recursive solutions for the risk-neutral prices of zero-coupon consumption claims. Let $P_{nt}^{c,rn}/C_t$ denote the risk-neutral price-dividend ratio of a zero-coupon claim on consumption at time $t+n$. The risk-neutral price-consumption ratio of a claim to the entire stream of future consumption equals:

$$\frac{P_t^{c,rn}}{C_t} = \sum_{n=1}^{\infty} \frac{P_{nt}^{c,rn}}{C_t}. \quad (\text{A136})$$

For $n \geq 1$, we guess and verify there exists a function $F_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1})$, such that

$$\frac{P_{nt}^{c,rn}}{C_t} = F_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}). \quad (\text{A137})$$

We start by deriving the analytic expression for F_1^{rn} . The one-period risk-neutral zero-coupon price-consumption ratio solves

$$\frac{P_{1,t}^{c,rn}}{C_t} = \mathbb{E}_t \left[M_{t+1}^{rn} \frac{C_{t+1}}{C_t} \right] \quad (\text{A138})$$

$$= \exp(-r_t + u_t) \mathbb{E}_t \left[\frac{C_{t+1}}{C_t} \right] \quad (\text{A139})$$

$$= \exp(-\gamma \mathbb{E}_t x_{t+1} + \gamma(\phi - \theta_1)x_t - \gamma\theta_2 x_{t-1} - \bar{r}) \mathbb{E}_t \left[\frac{C_{t+1}}{C_t} \right] \quad (\text{A140})$$

Using (A87) to substitute for consumption growth, we can derive the following analytic expression for f_1^{rn} :

$$f_1^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = g - \bar{r} - e_1[(\gamma\theta_1 - \gamma\phi + \phi)I - (1 - \gamma)B]A^{-1}\tilde{Z}_t - \gamma\theta_2 x_{t-1} + \frac{1}{2}\sigma_c^2. \quad (\text{A141})$$

Next, we solve for f_n , $n \geq 2$ iteratively:

$$\frac{P_{nt}^{c,rn}}{C_t} = \exp(-\gamma \mathbb{E}_t Y_{1,t+1} + \gamma(\phi - \theta_1)Y_{1,t} - \gamma\theta_2 x_{t-1} - \bar{r}) \mathbb{E}_t \left[\frac{C_{t+1}}{C_t} F_{n-1}^{rn}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right] \quad (\text{A142})$$

This gives the following expression for f_n^{rn} :

$$\begin{aligned} f_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \log \left[\mathbb{E}_t \left[\exp(-\gamma \mathbb{E}_t Y_{1,t+1} + \gamma(\phi - \theta_1)Y_{1,t} - \gamma\theta_2 x_{t-1} - \bar{r} \right. \right. \\ &\quad \left. \left. + g - \phi Y_{1,t} + \mathbb{E}_t Y_{1,t+1} + \sigma_c \epsilon_{1,t+1} \right. \right. \\ &\quad \left. \left. + f_{n-1}^{rn}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right) \right] \right]. \end{aligned} \quad (\text{A143})$$

Finally, we re-write $f_{n,t}^{rn}$ as an expectation involving $f_{n-1,t+1}^{rn}$, the state variables $\tilde{Z}_t$, and period $t+1$ shocks:

$$\begin{aligned} f_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \log \left[\mathbb{E}_t \left[\exp \left(e_1[(1 - \gamma)B - (\phi - \gamma(\phi - \theta_1))I]A^{-1}\tilde{Z}_t - \gamma\theta_2 x_{t-1} \right. \right. \right. \\ &\quad \left. \left. + g - \bar{r} + \sigma_c \epsilon_{1,t+1} + f_{n-1}^{rn}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t) \right) \right] \right]. \end{aligned} \quad (\text{A144})$$

B.5 Risk-neutral returns

We plug risk-neutral price-consumption ratios and bond prices into equations (A111) through (A117). This gives risk-neutral returns on the consumption claim, risk-neutral log excess

bond returns, and risk-neutral bond yields. We then substitute risk-neutral returns on the consumption claim into (A123)-(A124) to obtain risk-neutral log excess stock returns.

C Expected growth shock interpretation of preference shock

I now show that the preference shock for bonds ξ_t can be interpreted as a shock to growth expectations. I first show that this equivalence holds exactly when price and wage inflation are equal, and the EIS equals one, i.e. $\gamma = 1$. I then discuss how this intuition translates to the more general case and which additional assumptions would be needed.

Assume that expected productivity growth is subject to shocks that are conditionally homoskedastic and iid

$$E_t a_{t+1} = \nu + a_t + (1 - \phi)x_t + \frac{1}{\gamma}\xi_t, \quad (\text{A145})$$

that is $\frac{1}{\gamma}\xi_t$ is a shock to expected potential output. Expected consumption growth then equals

$$E_t \Delta c_{t+1} = E_t x_{t+1} - \phi x_t + \frac{1}{\gamma}\xi_t. \quad (\text{A146})$$

The consumption surprise in the SDF and its volatility therefore remain unchanged:

$$c_{t+1} - E_t c_{t+1} = x_{t+1} - E_t x_{t+1}, \quad (\text{A147})$$

$$\sigma_c^2 = e_1 \Sigma \Sigma_u M' e'_1. \quad (\text{A148})$$

The first-order asset pricing condition for the real risk-free rate takes the form

$$r_t = \gamma E_t \Delta c_{t+1} + \gamma \theta_1 x_t + \gamma \theta_2 x_{t-1}, \quad (\text{A149})$$

$$= \gamma E_t x_{t+1} - \gamma \phi x_t + \gamma \theta_1 x_t + \gamma \theta_2 x_{t-1} + \xi_t \quad (\text{A150})$$

Rearranging shows that ξ_t gives rise to a demand shock in the macroeconomic Euler equation with exactly the same functional form as in the main paper:

$$x_t = f^x E_t x_{t+1} + \rho^x x_{t-1} - \psi r_t + \underbrace{\frac{1}{\gamma(\phi - \theta_1)} \xi_t}_{v_{x,t}} \quad (\text{A151})$$

We now just need to check how an expected growth shock affects stocks and bonds. The real bond pricing recursion is given by

$$P_{n,t} = E_t [M_{t+1} P_{n-1,t+1}], \quad (\text{A152})$$

$$= E_t [\exp(-\xi_t - \gamma(x_{t+1} - \phi x_t) - \gamma \Delta s_{t+1}) P_{n-1,t+1}]. \quad (\text{A153})$$

The nominal bond pricing recursion similarly becomes

$$P_{n,t}^s = E_t [M_{t+1} P_{n-1,t+1}], \quad (\text{A154})$$

$$= E_t [\exp(-\xi_t - \gamma(x_{t+1} - \phi x_t) - \gamma \Delta s_{t+1}) P_{n-1,t+1}^s \exp(-\pi_{t+1})]. \quad (\text{A155})$$

This shows that the bond pricing recursion is exactly as in the main paper. Intuitively, higher expected potential output in the future shifts households' preferences away from saving in bonds at any level of current consumption.

The recursion for a zero-coupon consumption claim becomes

$$\begin{aligned} \frac{P_{n,t}^c}{C_t} &= E_t \left[M_{t+1} \frac{C_{t+1}}{C_t} \frac{P_{n-1,t+1}^c}{C_{t+1}} \right], \quad (\text{A156}) \\ &= \exp(-\xi_t + \frac{1}{\gamma} \xi_t) E_t \left[\exp(-\gamma(x_{t+1} - \phi x_t) - \gamma \Delta s_{t+1}) \exp(x_{t+1} - \phi x_t) \frac{P_{n-1,t+1}^c}{C_{t+1}} \right] \end{aligned}$$

In the special case where the EIS equals one (i.e. $\gamma = 1$) the term outside the expectations drops out and I recover the consumption claim recursion in the main paper. So, in this case, the expected growth shock ξ_t affects the prices of consumption claims only to the extent that it changes macroeconomic and surplus consumption dynamics, just as in the main paper. Intuitively, a shock to future expected growth has offsetting effects on the desire to substitute intertemporally and expected dividend growth. With sticky prices and flexible wages ξ_t can therefore exactly be interpreted as a shock to expected potential output.

Table A1 shows that setting $\gamma = 1$ and $\phi = 1$ closely replicates the macroeconomic and asset pricing properties in Table 2 in the main paper, so the model in the main paper closely resembles this special case where the growth expectations interpretation is exactly valid.

Generalizing this interpretation of ξ_t as an expected growth shock to the exact parameterization in the main paper would need to rely on two small modifications and, given the similarity of the results in Table A1, would likely leave the main model properties unchanged. I prefer $\gamma = 2$ simply because it gives a better fit for the output gap response to an identified monetary policy shock as in Pflueger and Rinaldi (2022), which is important to not overstate the power of monetary policy for the economy and risk premia. To carry through the interpretation as an expected growth shock to the case with $\gamma = 2$, as in the main paper, one

would need to assume that investors expect the expected growth shock to enter in a levered manner into the dividend growth of a consumption claim, i.e. $E_t \Delta d_{t+1} = \gamma \Delta c_{t+1}$, similar to the leverage assumption in [Abel \(1990\)](#). I do not pursue this model of leverage here, but it is well-known that this type of assumption increases the volatility of stock returns while keeping the Sharpe ratio unchanged, so it seems unlikely that the model properties would change in any meaningful way if I replaced the model of leverage in the main paper by this alternative one.

When wages are sticky and prices are flexible, the shock to growth expectations enters into the wage-price inflation relationship as follows:

$$E_t \pi_{t+1}^p = E_t \pi_{t+1}^w - (1 - \phi) \hat{x}_t - \frac{1}{\gamma} \xi_t \quad (\text{A157})$$

Introducing plausible price stickiness along with wage stickiness would scale down the last term close to zero, because wage inflation then enters into price inflation with a lag. This would then restore the interpretation that expected growth shocks leave all macroeconomic dynamics and asset price recursions fundamentally unchanged. However, because price and wage inflation are already very similar in the model, I do not pursue this complication here.

D Additional Model Results

D.0.1 Rational Inflation Expectations

Because the calibration procedure separately calibrates the adaptive inflation expectations parameter ζ holding all other parameters constant, it is important to check that the macroeconomic moments used to calibrate the volatilities of shocks and monetary policy parameters do not suffer unreasonably from changing ζ . Because the 2001.Q2-2019.Q4 calibration uses $\zeta = 0$ this is not a concern here. [Figure A1](#) shows the model-implied impulse responses against the data for the 1979.Q5-2001.Q1 calibration with $\zeta = 0$, analogously to [Figure 2](#) in the main paper.

Comparing against [Figure 2](#) in the main paper shows that most impulse responses are unaffected by changing ζ , and the only difference in the impulse responses with $\zeta = 0.6$ is in the long end of Panel B, i.e. the lead-lag relationship between the fed funds rate and inflation. With $\zeta = 0$, the inflation response in the model is naturally more persistent. While this less persistent inflation response superficially appears to fit the data better, I still prefer the higher adaptiveness parameter for several reasons. First, long-term inflation is measured with noise because of the substantial short-term volatility in inflation that is irrelevant for monetary policy. Second, consistent with a higher value for $zeta = 0.6$, a volatile persistent

component in inflation for the 1980s is in line with a long-standing econometrics literature that has decomposed inflation into its permanent and transitory components (Stock and Watson (2007)). Finally, partially adaptive inflation expectations are help match features in inflation expectations, such as the predictability of inflation forecast errors in Table 3 in the main paper.

Figure A2 further illustrates the mechanism by which the adaptiveness of inflation expectations acts on the economy and inflation. In particular, we see that the inflation response to a supply shock is much less persistent when $\zeta = 0$ than when $\zeta = 0.6$, as in Figure 3 in the main paper.

D.1 Robustness

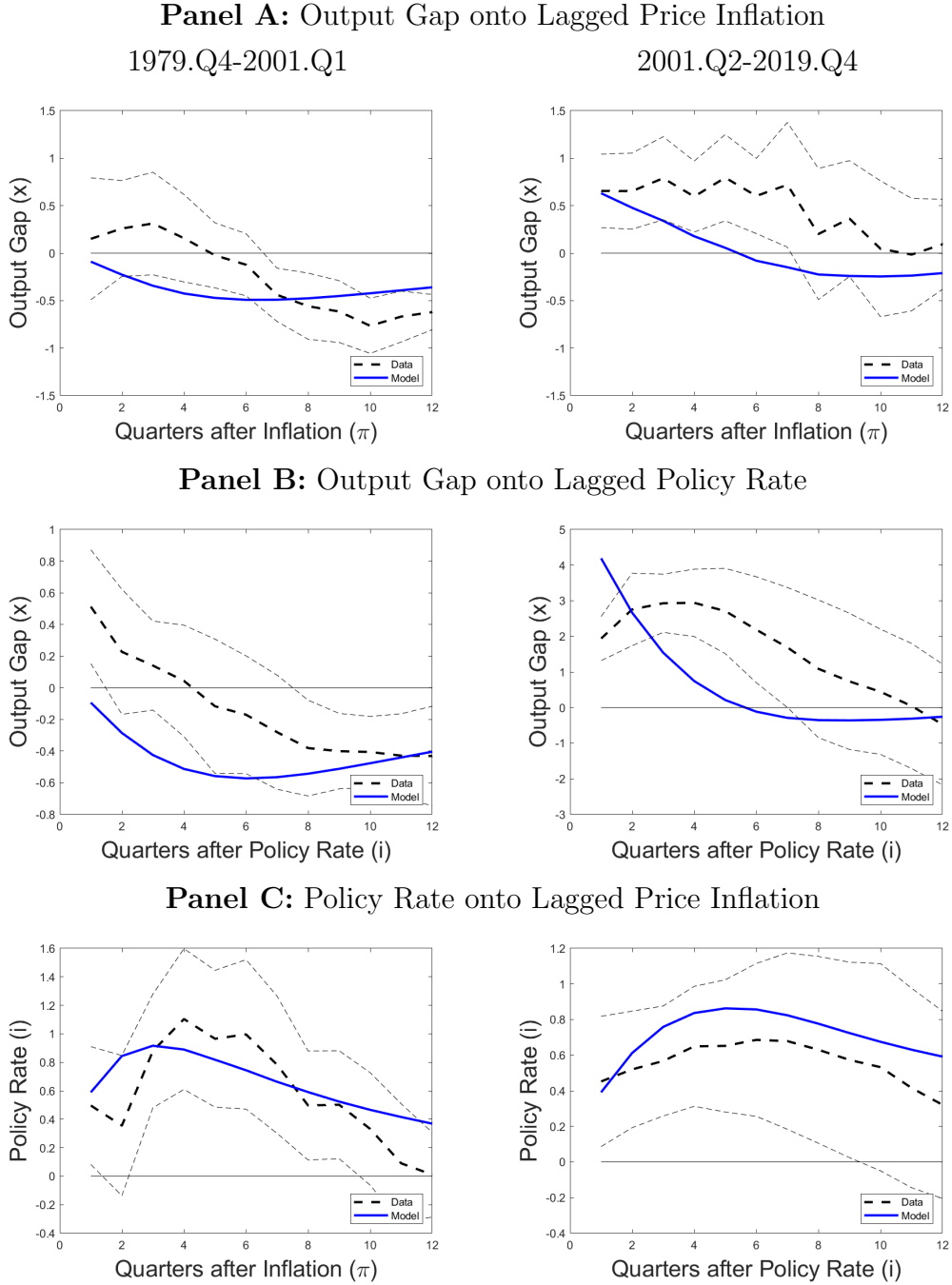
Table A1 shows the main model moments as in Table 2 of the main paper for some different parameter combinations, including the cases with $\gamma = 1$, $\phi = 1$ and $\zeta = 0$. Implied nominal bond-stock betas are positive for the 1980s calibration and negative for the 2000s calibration and the all moments are qualitatively or even quantitatively unchanged.

The case with $\phi = 1$ is of interest because this switches off the difference between price and wage inflation (equation (26) in the main paper. We see that setting $\phi = 1$ leaves all asset pricing and macroeconomic moments qualitatively and quantitatively unchanged. This tells us that the model properties are unaffected whether we assume sticky wages or sticky prices.

Changing the utility curvature parameter to $\gamma = 1$ implies a positive nominal bond-stock for the 1980s calibration and a negative nominal bond-stock beta for the 2000s calibration. The nominal bond-stock for the 1980s calibration is somewhat smaller in magnitude than in the baseline. While the Campbell-Shiller bond excess return predictability coefficient is still positive and within a 90% confidence interval of the empirical coefficient, it is also smaller. While this is not my preferred case, the the case with $\gamma = 1$ and $\phi = 1$ is of interest because it shows one extremely simple example where preference shocks for bonds are functionally isomorphic to shocks to expected potential output growth.

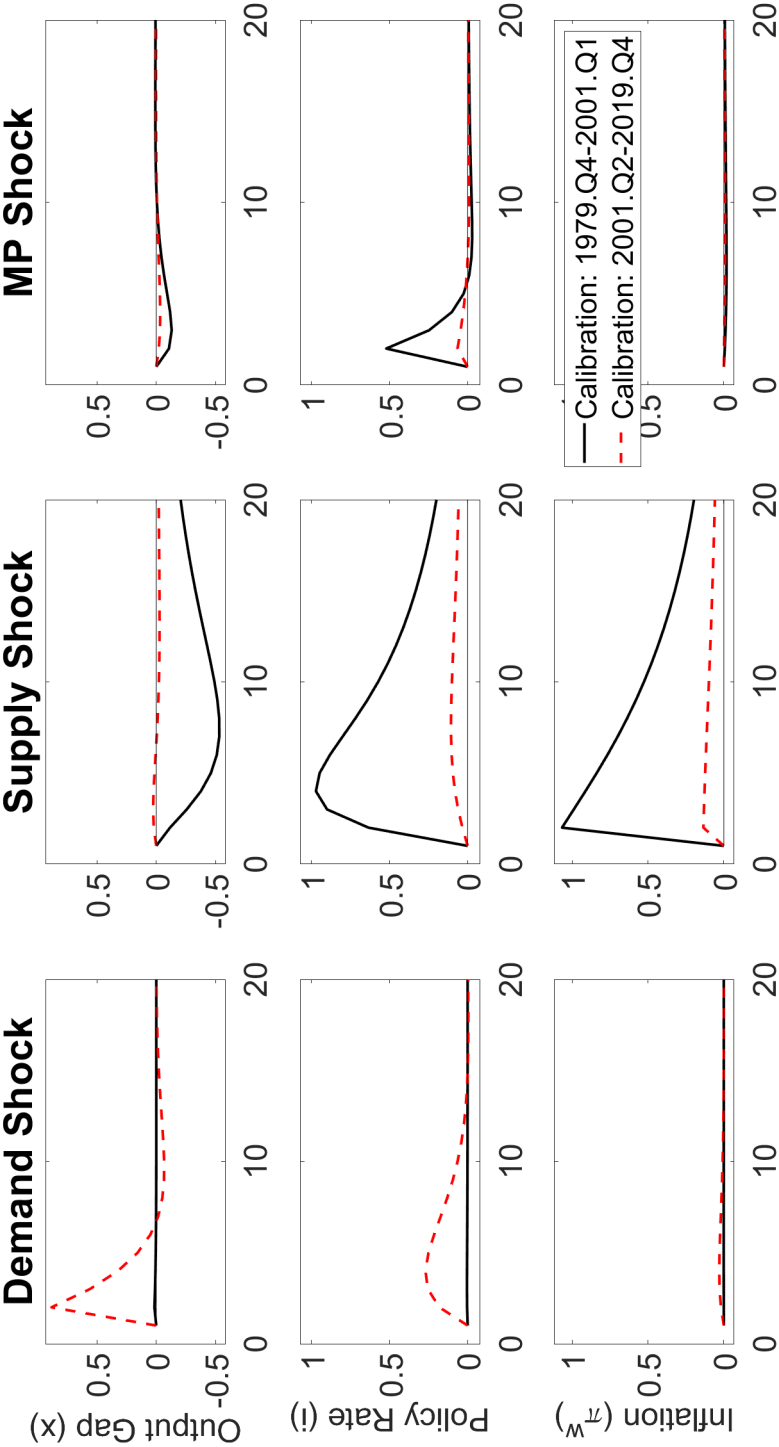
Setting $\zeta = 0$ (i.e. inflation expectations are rational) changes the Campbell-Shiller excess bond return predictability coefficient to zero for the 1980s calibration, consistent with the result in Figure 5 in the main paper, but leaves all other moments unchanged. This tells us that ζ is pinned down by the predictability of bond excess returns without affecting any of the other model implications.

Figure A1: Empirical Output Gap, Inflation, and Policy Rate Dynamics Pre- vs. Post-2001 with Rational Inflation Expectations



This figure is analogous to Figure 2 in the main paper, but it sets $\zeta = 0$ for both calibrations.

Figure A2: Model Macroeconomic Impulse Responses with Rational Inflation Expectations



This figure is analogous to Figure 3 in the main paper, but it sets $\zeta = 0$ for both calibrations.

Table A1: Model and Data Moments Robustness

Panel A: 1979.Q4-2001.Q1 Calibration					
	Baseline	$\phi = 1$	$\gamma = 1$	$\gamma = 1, \phi = 0$	$\zeta = 0$
Stocks					
Equity Premium	7.33	6.91	5.87	5.69	7.14
Equity Vol	14.95	13.95	15.16	14.43	14.40
Equity SR	0.49	0.50	0.39	0.39	0.50
AR(1) ρ	0.96	0.95	0.95	0.95	0.95
1 YR Excess Returns on ρ	-0.38	-0.43	-0.41	-0.43	-0.41
1 YR Excess Returns on ρ (R2)	0.06	0.07	0.05	0.05	0.06
Bonds					
Yield Spread	2.28	2.20	0.45	0.44	2.45
Return Vol.	15.82	15.39	12.89	12.59	10.12
Nominal Bond-Stock Beta	0.86	0.85	0.42	0.37	0.65
Real Bond-Stock Beta	0.05	0.06	0.05	0.05	0.09
1 YR Excess Returns on slope	1.26	1.24	0.25	0.24	-0.36
1 YR Excess Returns on slope (R2)	0.01	0.01	0.00	0.00	0.00
Macroeconomic Volatilities					
Std. Annual Cons. Growth	0.76	0.73	0.89	0.87	0.98
Std. Annual Change Fed Funds Rate	1.64	1.63	1.62	1.62	2.15
Std. Annual Change 10-Year Subj. Infl. Forecast	0.73	0.71	0.64	0.62	0.87

This table shows robustness results for the calibration moments shown in Table 2 in the main paper. Panel A sets all parameters to their 1979.Q4-2001.Q1 calibration values and changes the indicated parameters one at a time. Panel B sets all parameters to their 2001.Q2-2019.Q4 calibration values and changes the indicated parameters one at a time.

Model and Data Moments Robustness (continued)

Panel B: 2001.Q2-2019.Q4 Calibration

	Baseline	$\phi = 1$	$\gamma = 1$	$\gamma = 1, \phi = 0$	$\zeta = 0.6$
Stocks					
Equity Premium	9.15	9.15	5.69	5.68	9.15
Equity Vol	19.29	19.29	15.69	15.68	19.29
Equity SR	0.47	0.47	0.36	0.36	0.47
AR(1) ρ	0.93	0.93	0.95	0.95	0.93
1 YR Excess Returns on ρ	-0.38	-0.38	-0.32	-0.32	-0.38
1 YR Excess Returns on ρ (R2)	0.14	0.14	0.10	0.10	0.14
Bonds					
Yield Spread	-0.58	-0.60	-0.24	-0.25	-0.62
Return Vol.	2.12	2.16	1.42	1.44	2.62
Nominal Bond-Stock Beta	-0.09	-0.09	-0.07	-0.07	-0.10
Real Bond-Stock Beta	-0.08	-0.08	-0.07	-0.07	-0.08
1 YR Excess Returns on slope	-0.31	-0.31	-0.16	-0.15	-0.39
1 YR Excess Returns on slope (R2)	0.01	0.01	0.00	0.00	0.00
Macroeconomic Volatilities					
Std. Annual Cons. Growth	1.59	1.62	1.44	1.47	1.58
Std. Annual Change Fed Funds Rate	0.65	0.68	0.56	0.58	0.61
Std. Annual Change 10-Year Subj. Infl. Forecast	0.12	0.12	0.10	0.10	0.09

This table shows robustness results for the calibration moments shown in Table 2 in the main paper. Panel A sets all parameters to their 1979.Q4-2001.Q1 calibration values and changes the indicated parameters one at a time. Panel B sets all parameters to their 2001.Q2-2019.Q4 calibration values and changes the indicated parameters one at a time.

References

- Abel, Andrew B (1990) “Asset Prices under Habit Formation and Catching up with the Joneses,” *American Economic Review*, 38–42.
- Campbell, John Y and John H Cochrane (1999) “By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior,” *Journal of Political Economy*, 107, 205–251.
- Campbell, John Y and Sydney Ludvigson (2001) “Elasticities of Substitution in Real Business Cycle Models with Home Production,” *Journal of Money, Credit and Banking*, 33 (4), 847–875.
- Campbell, John Y, Carolin Pflueger, and Luis M Viceira (2020) “Macroeconomic drivers of bond and equity risks,” *Journal of Political Economy*, 128 (8), 3148–3185.
- Christiano, Lawrence H., Martin Eichenbaum, and Charles L. Evans (2005) “Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy,” *Journal of Political Economy*, 113 (1), 1–45.
- Greenwood, Jeremy, Zvi Hercowitz, and Gregory W Huffman (1988) “Investment, capacity utilization, and the real business cycle,” *American Economic Review*, 402–417.
- Lucas, Robert E. Jr. (1988) “On the Mechanics of Economic Development,” *Journal of Monetary Economics*, 22, 3–42.
- Pflueger, Carolin and Gianluca Rinaldi (2022) “Why does the fed move markets so much? a model of monetary policy and time-varying risk aversion,” *Journal of Financial Economics*, 146 (1), 71–89.
- Rotemberg, Julio J (1982) “Monopolistic price adjustment and aggregate output,” *Review of Economic Studies*, 49 (4), 517–531.
- Smets, Frank and Rafael Wouters (2007) “Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach,” *American Economic Review*, 586–606.
- Stock, James H and Mark W Watson (2007) “Why has US inflation become harder to forecast?,” *Journal of Money, Credit and banking*, 39, 3–33.